Updated 9/9/2026

GEO/SUD 3/430:
Sustainable Urban Transportation
Fall Quarter 2027
Tuesdays 6pm-9:15pm
Online

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Office hours: By appointment

COURSE DESCRIPTION

Over the past several decades, the capacity of urban areas to provide transportation that sustains our planet's well-being has come under increasing strain. As cities work to develop more sustainable practices, conventional transportation thinking, design, and policy are being reevaluated. This online course examines tools and strategies for addressing the societal, economic, and environmental challenges facing urban transportation today. Students will explore current issues in the field and the approaches transportation planners, engineers, urban designers, and policymakers use to address them. Throughout the quarter, students will apply the conceptual and technical tools they learn to a Chicago-based transportation project.

COURSE OBJECTIVES/LEARNING OUTCOMES

This course is designed to examine key aspects of sustainable transportation practice today, including:

- **Critically evaluate** the evolution, benefits, and challenges of different transportation modes through course readings and discussions;
- **Apply** analytical tools (e.g., RStudio) and procedures (e.g., spatial analysis) to assess transportation safety, equity, mobility, and accessibility;
- **Envision** how society can incorporate emerging forms of urban mobility, design, and planning to advance principles of sustainability; and
- **Collaborate** in groups to draft a sustainable transportation design, policy, and/or plan for presentation to the class.

REQUIRED TEXTS

You are not required to purchase a text for this course. All required and supplemental readings (i.e., book chapters, journal articles, professional reports, plans and designs) will be made available via D2L or in class.

ASSESSMENT/GRADING

Student performance will be evaluated with regard to four categories of activities. The breakdown of total points by activity is shown in the table below.

Activity	Points (undergrad)	Points (grad)
In-class assignment	10 @ 2 = 20 points	10 @ 2 = 20 points
Methods exercise	4 @ 15 = 60 points	4 @ 15 = 60 points
Group presentation	1 @ 20 = 20 points	1 @ 20 = 20 points
Travel log and report		1 @ 10 = 10 points
Total	***100 points***	***110 points***

Letter Grade	% of points earned
A	93 or Higher
A-	90-92
B+	87-89
B	83-86
B-	80-82
C+	77-79
C	73-76
C-	70-72
D+	67-69
D	63-66
F	62 or Less

ACTIVITIES

Throughout the quarter, students will be evaluated on three categories of activities: (1) in-class assignments, (2) methods exercises, and (3) group presentation. To participate effectively in this class, plan to spend an average of 3 to 5 hours per week on course activities in addition to attending class. Brief descriptions of the three activities follow below; please review the associated rubrics on the D2L website for detailed information on how each activity will be graded.

(1) **In-class assignments**: Each week, students are expected to attend class, think critically, and actively engage with concepts presented during class time and in the required readings. In-class assignments such as task-based group discussions, quizzes, or other activities will be assigned weekly and take approximately 20–30 minutes to complete.

(2) **Methods exercises**: Students will complete methods exercises that apply GIS tools and techniques commonly used in transportation planning, engineering, and policy analysis. Four such exercises are required over the course of the quarter.

(3) **Group presentation**: Working in small groups of 3–4, students will design and produce a presentation that explores an instructor-approved transportation-related problem and proposes one or more sustainability-informed approaches or solutions. Groups will present their work to the class in a 15-minute presentation during finals week.

TOPICAL CLASS AGENDA

- Week 1 (9/15): Class orientation; Foundational concepts (Exercise #1 assigned)
- Week 2 (9/22): Walking and the value of slow transport
- Week 3 (9/29): Bicycling and active transportation (Exercise #1 due; Exercise #2 assigned)
- Week 4 (10/6): Transit and the efficiencies of mass movement (Rough project proposal due)
- Week 5 (10/13): Driving and the implications of automation (Exercise #2 due)
- Week 6 (10/20): Reclaiming urban waterways (Exercise #3 assigned)
- Week 7 (10/27): Just transportation (Refined project proposal)
- Week 8 (11/3): Transportation and public health (Exercise #3 due; Exercise #4 assigned)
- Week 9 (11/10): Economic benefits of sustainable transportation
- Week 10 (11/17): Advocating for sustainable transportation systems (Exercise #4 due)
- Finals week (11/24): (Group project presentations)
- Final projects, outstanding assignments due Friday, 11/27 by end of day

DETAILED CLASS AGENDA

Part One: Introduction to sustainable urban transportation. Course formalities; Key concepts

- Week One: *Foundational concepts in sustainable urban transportation*
    - Schneider, Robert J. 2013. “Theory of Routine Mode Choice Decisions: An Operational Framework to Increase Sustainable Transportation.” Transport Policy 25. 128–37.
    - **Methods exercise #1 prep:** Analyze transportation mode share and pedestrian fatalities for a student-selected county in the United States.

Part Two: Modes of transportation. Weeks two through five are organized around a primary mode of transportation. We discuss the character of the various modes, their evolution in planning and relationships with society and patterns of physical development.

- Week Two: *Walking and the value of slow transport.*
    - **Required reading**: Speck, Jeff. 2018. Walkable City Rules: 101 Steps to Making Better Places. Washington, D. C., UNITED STATES: Island Press.

- Week Three: *Bicycling and active transportation*
    - **Required reading:** Dill, Jennifer, and Nathan McNeil. “Four Types of Cyclists?: Examination of Typology for Better Understanding of Bicycling Behavior and Potential.” Transportation Research Record 2387, no. 1 (January 1, 2013): 129–38.
    - **Methods exercise #2 prep:** Analyze usage of Divvy bikeshare program across Chicago neighborhoods by sociodemographic group.
    - **Methods exercise #1 due**

- Week Four: *Transit and the efficiencies of mass movement*
    - **Required reading:** Walker, Jarrett. Human Transit: How Clearer Thinking about Public Transit Can Enrich Our Communities and Our Lives. 1st Edition. Washington, DC: Island Press, 2011.

- Week Five: *Driving and the implications of automation*
  - **Required reading:** Townsend, Dr Anthony. 2014. "Re-Programming Mobility: The Digital Transformation of Transportation in the United States." New York University.
  - **Methods exercise #3 prep:** Analyze public transit data pre- and post-COVID.
  - **Methods exercise #2 due**

- Week Six: *Reclaiming urban waterways for recreational active transportation*
  - **Required reading:** Cohen, Adam, and Susan Shaheen. 2016. "Planning for Shared Mobility." PAS-583. Planning Advisory Service. American Planning Association.

Part Three: Analytical dimensions of sustainable urban transportation. Weeks seven through ten explore analytical frameworks that can be used to evaluate and communicate transportation system performance.

- Week Seven: *Just transportation*
  - **Required reading:** Hananel, Ravit, and Joseph Berechman. "Justice and Transportation Decision-Making: The Capabilities Approach." Transport Policy 49 (July 2016): 78–85.
  - **Methods exercise #3 due**
  - **Methods exercise #4 prep:** Transportation performance and public health outcomes.

- Week Eight: *Transportation, the environment and public health*
  - **Required reading:** Younkin, Samuel G., Henry C. Fremont, and Jonathan A. Patz. "The Health-Oriented Transportation Model: Estimating the Health Benefits of Active Transportation." Journal of Transport & Health 22 (September 1, 2021)

- Week Nine: *The economic benefits of sustainable urban transportation*
  - **Methods exercise #4 due**

- Week Ten: *Advocating for sustainable transportation systems*
  - **Required reading:** OECD. "Chapter 3 Managing Transportation Demand: Offering Attractive Choices." In ITF Transport Outlook 2023. Paris: Organisation for Economic Co-operation and Development, 2023.

Part Four. Final presentation of group projects, submission of reports
  - **Group project presentation**
  - **Group report due**

COURSE POLICIES

Academic Integrity

Students are expected to behave in a manner consistent with DePaul's Code of Student Responsibility. Academic honesty and integrity are expected at all times. Academic dishonesty, such as cheating or copying during exams, will be punished severely. Plagiarism–using someone else's work without acknowledgment and, therefore, presenting their ideas or quotations as your own work–is strictly forbidden. DePaul University officials will be informed of any instance of academic dishonesty and notification will be placed in your file. Please read the DePaul Academic Integrity Resources page (http://offices.depaul.edu/oaa/faculty-resources/teaching/academic-integrity/Pages/default.aspx) for definitions and explanations of plagiarism and the University's Academic Integrity expectations for students. Cutting and pasting text taken directly from a website (including AI) without appropriate referencing and quotation marks is plagiarism and is forbidden. Submitting work that has any part cut and pasted directly from the internet is grounds for an automatic grade of zero.

Disability Accommodations

Students seeking disability-related accommodations are required to register with DePaul's Center for Students with Disabilities (CSD). They will provide you with the accommodations and support services needed to ensure your success. You can reach CSD on the Loop Campus at (312) 362-8002 and CSD on the Lincoln Park Campus at (773) 325-1677. Students are invited to contact me privately to discuss your challenges and how I may assist in facilitating the accommodations you will use in this course.

Writing Resources and Expectations

The University Center for Writing-Based Learning (Writing Center) provides help free-of-charge to all members of the DePaul University community, including students, alumni, faculty, and staff. Writing Center tutors are trained to approach tutorials as your peers, acting as engaged readers of your written work. To schedule an appointment, visit www.depaul.edu/writing.

Social, Cultural, and Behavioral Inquiry

This course satisfies the Social, Cultural, and Behavioral Inquiry domain requirement. Courses in this domain focus on the mutual impact of society and culture on individuals, and of individuals on society and culture. Particular attention is given to human relationships and behavior as they are influenced by social, economic, and political institutions; spatial and geographical factors; and the events and social and cultural forces at play in the contemporary world. The domain emphasizes the pursuit of knowledge through the development of theory and empirical investigation of the contemporary world. Courses in the domain explore such particular issues as poverty and economic opportunity, the environment, nationalism, racism, individual alienation, gender differences, and the bases of conflict and consensus in complex, urban societies and in global relations. After completing this course, students will be able to:

- Use the constructs of power, diversity, and/or culture to describe examples of where, why, and how inequities exists in modern society.
- Frame a theory about the relationship between individuals and modern society.
- Analyze central institutions and/or underlying social structures and their impact on the larger society.
- Articulate an argument based on theory and empirical evidence regarding the modern world.
- Analyze critically research and arguments about the modern world.
- Reflect, in writing, upon their role in the modern world, including their relationship to their own and/or other communities.
- Analyze social problems and public policies on the basis of ethics and values.